
Intelligence Engineering

Solving Real World Problems At Scale

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LECTURE X

Infrastructure and Tooling for MLOps

This lecture in one slide

X Infrastructure and Tooling for MLOps

Storage and Compute

Development Environment

Resource Management

ML Platform

Build Versus Buy

*why this lecture matters, and what
you can do after it*

1

Storage and Compute

Public Cloud Versus Private Data Centers

Public cloud

Private data centers

2

Development Environment

Dev Environment Setup

Dev environment

Standardizing Dev Environments

$P(y | x)$

[1]

From Dev to Prod: Containers

Containers

3

Resource Management

Cron, Schedulers, and Orchestrators

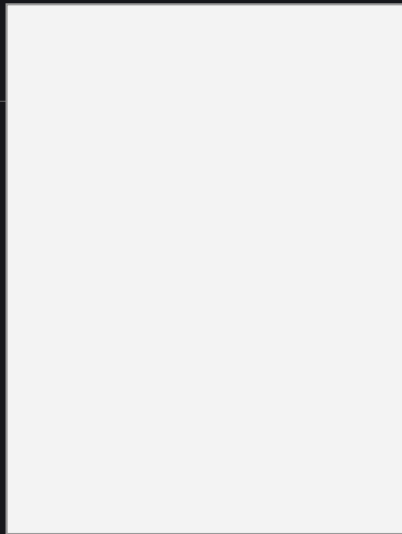
Cron

Schedulers

Orchestrators

Data Science Workflow Management

Data science workflow



$P(y | x)$

[1]

4

ML Platform

Model Deployment

$P(\hat{y} | x)$

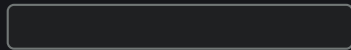
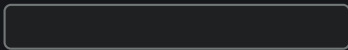
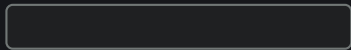
[1]

Model Store

Definition: Model store

Feature Store

$P(\hat{y} | x)$



[1]

5

Build Versus Buy

Build Versus Buy

$P(y|x)$

Build

Buy

[1]

6

Summary

Summary

$P(\hat{y} | x)$

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[1]

References

- 1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.